

Learning objectives

In this lesson you will learn to:

- Define what is meant by the term 'embedded system'
- Explain how an embedded system differs from a general-purpose computer
- Identify hardware and software components of embedded systems

For more information and additional student activities see Topic 3.1 of the student book.

Computer Systems

Previously, you have learned what makes up a computer.

What are the components required for a device to be considered a computer?

- Processor
- Memory
- Input and output.

General-purpose computers

There are two general categories of computers.

General-purpose computers are the ones that you are most familiar with. You would recognise them as computers.

Devices that could be considered general purpose include desktop PCs, laptops, smartphones and tablets.



Computers that don't look like computers

The vast majority of devices that could be considered a computer don't look anything like one.



Embedded systems

An embedded computer is usually a fairly simple computer, often on a single printed circuit board. It is usually responsible for a dedicated task.

Traffic lights are an example. The controller is usually out of sight but it has parts that:

- switch the lights
- keep track of time in each state
- receive input from the pedestrian button
- receive input from the road sensors, etc.



Characteristics of an embedded system

Let's identify some devices that use embedded systems.

Looking at the devices you have researched, what are some of the common characteristics of embedded systems?

- Task specific (single purpose)
- Low power consumption
- Small physical size
- Low maintenance
- Low cost
- Often real-time or responsive to their environment.

Hardware in an embedded system



Software in an embedded system

All computing devices need some kind of program to control them.

Generally, embedded computers have a small storage device to store their simple programs. As most embedded computers are single-purpose, their programs are short and simple, relying only on the fixed inputs and outputs of the system.

This means that programs written for embedded systems are often very small and they do not need to be modified once installed on the device. This is often known as **firmware**.

Wrap up: you have learned how to...

- ✓ Define what is meant by the term 'embedded system'
 - A small, single purpose computing device contained within and usually controlling a larger system
- ✓ Explain how an embedded system differs from a general-purpose computer
 - Generally has one specific purpose. Has fixed hardware components and firmware
- ✓ Identify hardware and software components of embedded systems
- ✓ Describe applications of embedded systems.